



Lai Weng Hong

Software Engineer

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Petaling Jaya, Selangor

My Portfolio Website

Weng Hong Lai | LinkedIn

Education

Universiti Malaysia Sarawak
Bachelor of Computer Science with Honours • 3.56/4.00
Software Engineering
Kota Samarahan, Sarawak • Oct 2020 - July 2024

Skills

- Cloud
 - AWS Lambda, AWS EventBridge, AWS DynamoDB, AWS Api Gateway, AWS Serverless Application Model (SAM), AWS Relational Database Service (RDS), Akamai Linode (CUDA), Docker
- Backend
 - Node.js, Laravel, .NET Core (C#), Express.js, Redis, MySQL
- Frontend
 - Vue.js 3, React (Vite), Tailwind CSS

Awards

NASA Space Apps Sarawak 2023 Hackathon - Senior Category Champion
20 Oct 2023
NASA Space Apps Sarawak
View News Article

Software Engineer II specializing in **event-driven serverless architectures** and **AI-powered media pipelines**. Experienced in designing decoupled, choreography-based systems using AWS (SAM, EventBridge, DynamoDB) and building full-stack B2B platforms with Laravel and Vue. Focused on optimizing cloud infrastructure for high-concurrency workloads and reducing operational costs through strategic architectural patterns.

Experience

Appsploration Sdn Bhd (Subsidiary of Innity) Petaling Jaya
Sept 2024 – Present
Software Engineer II Jan 2026 – Present
MomentsAI

- End-to-End Ownership:** Only developer responsible for the full lifecycle of the platform, moving from a local CLI tool to a production-ready AI video service.
- Infrastructure Migration:** Independently transitioned AI inference workloads from local CPU environments to **Akamai Linode GPU nodes (CUDA)**, enabling high-concurrency video processing for viral content generation.
- AI Pipeline Engineering:** Developed a unified "speech-to-face" synchronization pipeline by orchestrating **Whisper-1**, **Pyannote**, and **Buffalo_L** for automated diarization and face recognition.
- Queue Management:** Used **BullMQ** and **Redis** to handle asynchronous video rendering, preventing system timeouts during heavy media processing.

Software Engineer I Sept 2024 – Dec 2025
RewardsFX

- Sole Backend Architect:** Owned the full implementation of the RewardsFX backend, refining an initial roadmap into a high-scale reactive loop using DynamoDB Streams.
- Reactive Loop Design:** Refined the initial project roadmap into a fully decoupled system using **EventBridge → Lambda → DynamoDB Streams**. This allows 12+ services to react to data changes without blocking each other.
- Cost Optimization:** Reduced AWS costs and latency by connecting **API Gateway directly to EventBridge**, cutting out unnecessary Lambda executions for event routing.
- Idempotency & Data Integrity:** Developed a reusable **Lambda Layer** for distributed idempotency, guaranteeing 100% transaction accuracy across multiple reward types and ledger systems.
- DB Performance:** Optimized DynamoDB using **GSI/LSI indexes**, eliminating table scans and achieving sub-second response times for high-traffic lookups.
- Collaboration:** Led technical discussions with other department leads to align the backend architecture with business requirements.

Fusion Studio

- Asset Automation:** Extended the Laravel backend to support **Adobe Animate** assets; wrote logic to automate ZIP parsing and tracking script injection.
- Interactive Formats:** Built and deployed interactive ad templates using **Inertia.js** and **Vue 3**.

Datasonic Smart Solutions Sdn Bhd (Subsidiary of NexG) Putrajaya, Malaysia
Software Engineer (Part-Time & Intern) Feb 2023 - Oct 2023

- Project Kick-off (PERKESO):** Started the development of the **ARTs platform** from scratch, setting up the initial .NET Core project structure and backend logic.
- Security & Auth:** Built the login system using **JWT** and a **Singleton-pattern OTP engine** to manage secure, consistent token generation.
- Full-Stack Development:** Translated **Figma designs** into responsive components using **Quasar (Vue.js)** and integrated them with **.NET Core APIs**.
- Database Management:** Managed schemas and wrote optimized queries in **MySQL** to handle large-scale government records for the i-KAD system.
- Consultation:** Participated in design sessions with the Lead Engineer and external developers to finalize API specs and security standards.

Projects

i-KAD (Immigration Department of Malaysia) Feb 2023 - Oct 2023
A national identity management platform used to process, verify, and issue secure identification cards for foreign nationals in Malaysia.

PERKESO ARTs (Automated Reporting Tools) Aug 2023 - Oct 2023
An enterprise internal system built for SOCSO to automate the tracking of social security data and the generation of welfare claim reports.

RewardsFX Sept 2024 – Present
A centralized, multi-tenant loyalty infrastructure designed as the "Single Source of Truth" for reward logic and client engagement data across the entire organization.

Fusion Studio April 2025 - Present
A B2B SaaS platform used by major publishers to generate and deploy production-ready rich-media ads.

MomentsAI Nov 2025 - Present
An end-to-end AI platform that automates the extraction, transcription, and rendering of viral short-form clips from long-form video content.